Trainings

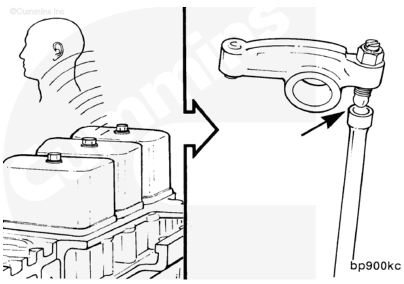
General Information

Overhead setting is only required at the interval specified in the appropriate Operation and Maintenance Manual/Owners Manual or when engine repairs cause removal of the rocker levers and/or loosening of the adjusting screws.

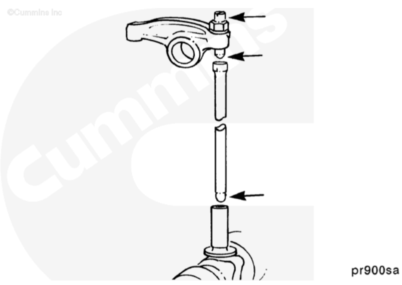
Diagnosing Component Malfunctions - Rocker Lever, Valve Stem, Push Rod, Tappet, and Camshaft.

Each cylinder of the engine has a separate rocker lever assembly. The pedestal support has drillings to route the oil flow to the shaft and levers.

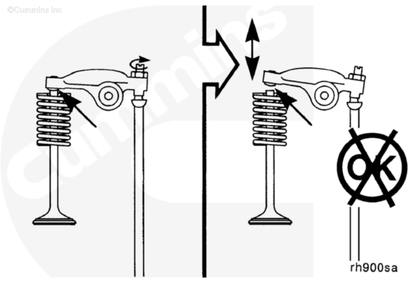
The rocker levers are push rod actuated and use an adjusting screw to control the clearance between the rocker lever and valve stem (crossheads for four four valves per cylinder engines). The rocker levers do **not** use a bushing in the bore for the rocker lever shaft. The rocker lever **must** be replaced if the bore is damaged or worn beyond the specification limit.



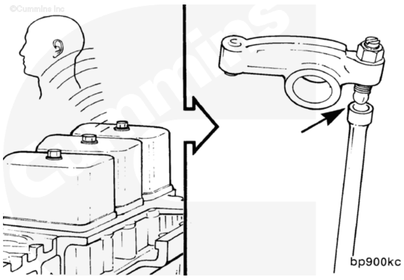
The ball end of the push tube fits into the ball socket in the tappet. The other end of the push rod has a ball socket in which the ball end of the rocker lever adjusting screw operates.



Excessive valve lash can indicate a worn valve stem, crosshead (for four valves per cylinder engines), push rod, valve tappet, or rocker lever.

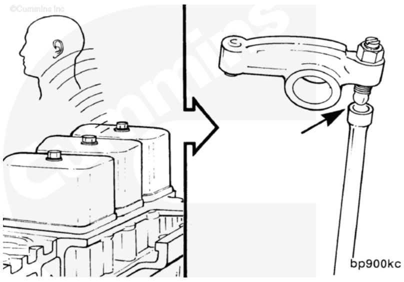


Loose rocker levers and the need to reset the valve clearance frequently can also indicate camshaft lobe or tappet wear. If an inspection of the rocker levers, valve stems, crosshead (for four valves per cylinder engines), and push rods does **not** show wear, then tappet and/or camshaft lobe wear can be suspected.



Refer to Procedures 001-008 (/qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html) and 004-015 (/qs3/pubsys2/xml/en/procedures/40/40-004-015-tr.html).

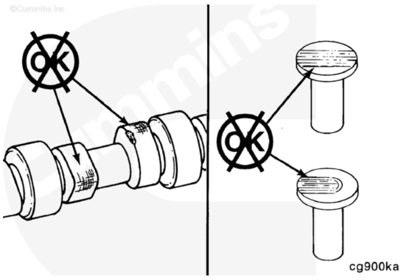
Loose rocker levers and the need to reset the valve clearance frequently can also indicate camshaft lobe or tappet wear. If an inspection of the rocker levers, valve stems, crosshead (for four valves per cylinder engines), and push rode does **not** show wear, then tappet and/or camshaft lobe wear can be suspected.



Contact a Cummins Authorized Repair Location.

⚠ CAUTION ⚠

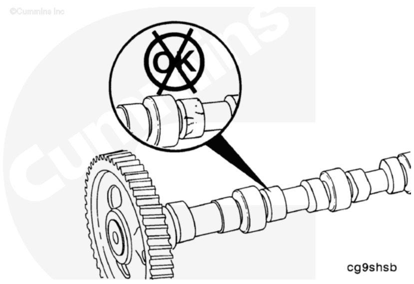
Anytime a new camshaft is installed, new tappets and push tubes must also be installed. Failure to do so can cause severe engine damage.



The camshaft lobes can be inspected after removing the lubricating oil pan.

The tappets can also be inspected with the lubricating oil pan removed. Remove the push rods, lift the tappets, and inspect the tappet faces.

Severely damaged camshaft journal(s) can generate metal chips that will be found in the lubricating oil pan and oil filter.

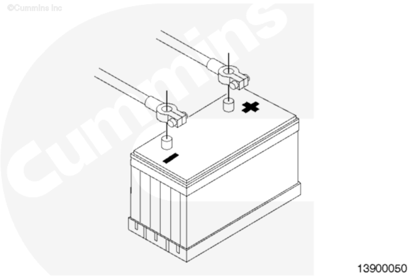


Note : As the clearance between the camshaft bushing(s) and camshaft journal(s) increase, oil pressure and volume will decrease, causing damage to the camshaft and tappets.

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

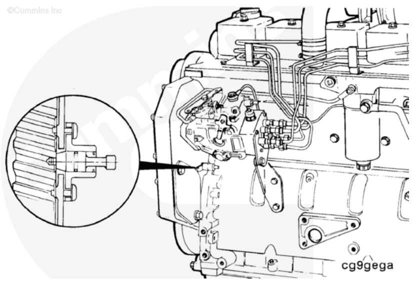


- Disconnect the batteries.

- Remove the rocker lever cover and gasket. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html).

Note : The timing pin is used to accurately locate TDC for setting the overhead. The timing pin is typically located below the fuel pump.

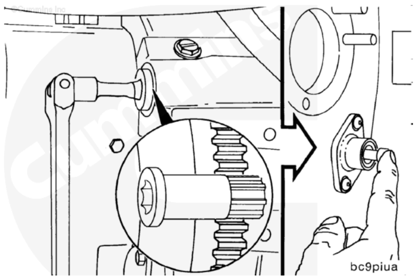
- for front gear train engines, in the front gear housing (shown)
- for rear gear train engines, in the rear gear housing (not shown)



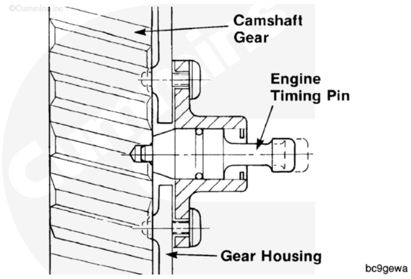
Locate top dead center (TDC) for cylinder number 1 by barring the crankshaft slowly while pressing on the engine timing pin. Barring the engine is recommended from the flywheel on the rear of the engine.

Note : Engine coolant temperature must be less than 60°C [140°F].

Using the barring tool Part Number 3824591, rotate the crankshaft slowly while pressing on the engine timing pin to locate TDC for cylinder number 1.

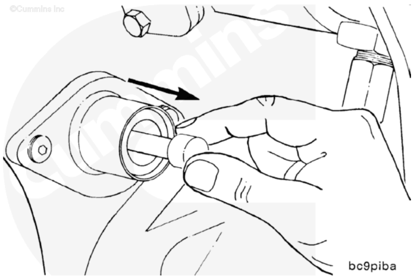


When the timing pin engages in the hole in the camshaft gear, cylinder number 1 is at TDC on the compression stroke.



⚠ CAUTION ⚠

To reduce the possibility of engine or timing pin damage, you must disengage the timing pin after locating top dead center.

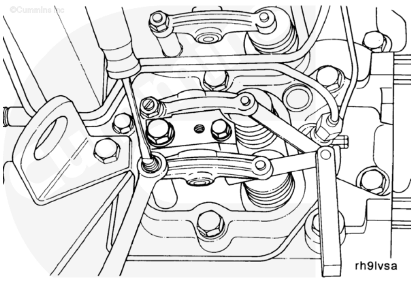


Adjust

B3.9, B5.9, and B4.5 Engines

Note : The clearance is correct when some resistance is "felt" when the feeler gauge is slipped between the valve stem and the rocker lever.

Note : Caution must be used when setting the exhaust valve lash on marine cylinder heads with rotators. The top of the valve stem is slightly recessed below the top of the valve rotator.



Measurements		
	mm	in
Intake Clearance:	0.254	0.010

Measurements		
	mm	in
Exhaust Clearance:	0.508	0.020

Four-Cylinder Engine Adjustment

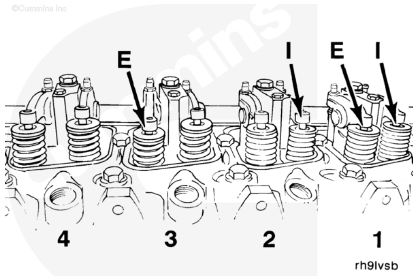
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** valves indicated by the arrows (E = exhaust, **I** = intake). Do **not** set valves that are **not** indicated.

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

Tighten the locknut and measure the valve lash again.

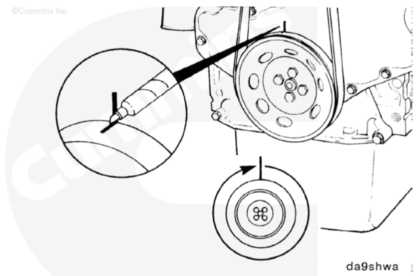
Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure the timing pin is disengaged.

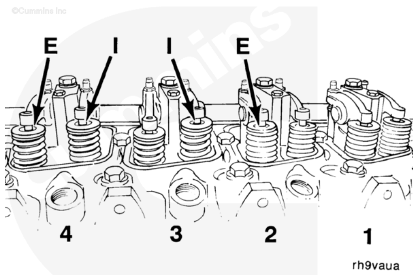
Mark the vibration damper and rotate the crankshaft 360 degrees.



Adjust the valves as indicated in the illustration.

Torque Value: 24 n•m [18 ft-lb]

Set **only** valves indicated by the arrows (E = exhaust, **I** = intake). Do **not** set valves that are **not** indicated.



Six-Cylinder Engine Valve Adjustment

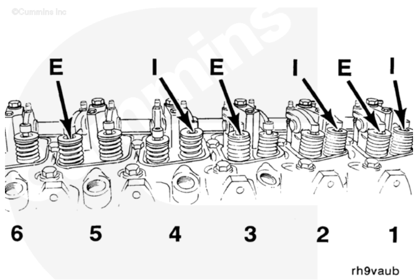
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, **I** = intake).

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

Tighten the locknut, and measure the valve lash again.

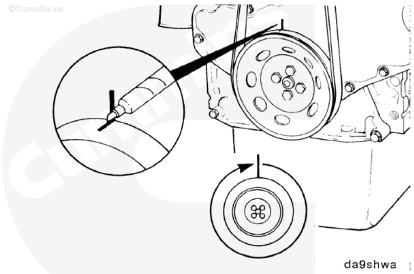
Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure timing pin is disengaged.

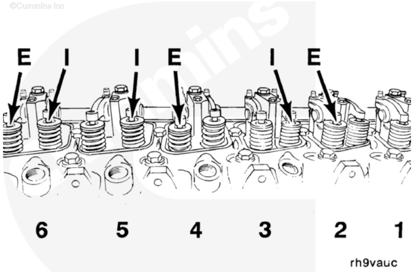
Mark the pulley, and rotate the crankshaft 360 degrees.



Adjust the valves as indicated in the illustration.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.

Torque Value: 24 n•m [18 ft-lb]



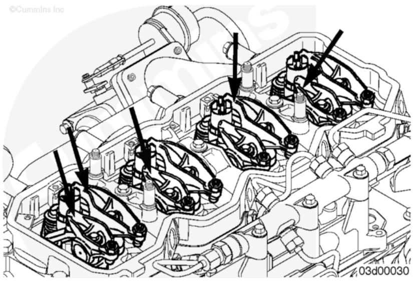
B4.5 RGT Engines

With the engine in this position, lash can be measured on the following rocker levers:

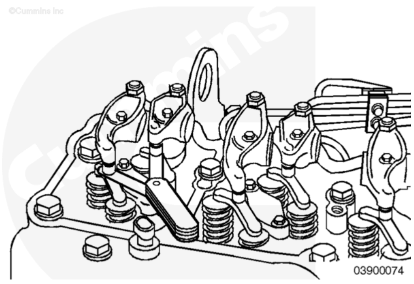
(E = exhaust, I = Intake)

Four-cylinder 1I, 1E, 2I, and 3E:

Six-cylinder 1I, 1E, 2I, 3E, 4I, and 5E.



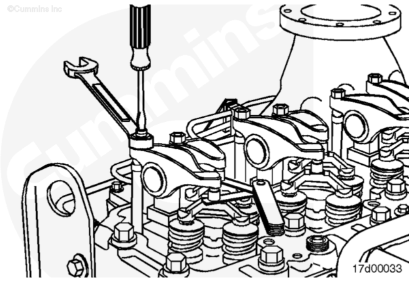
	mm		in
Intake	0.152	MIN	0.006
	0.381	MAX	0.015
Exhaust	0.381	MIN	0.015
	0.762	MAX	0.030



Note : Checking the overhead setting is usually performed as part of a troubleshooting procedure, and resetting is not required during checks as long as the lash measurements are within the above ranges.

Note : The clearance is correct when some resistance is "felt" when the feeler gauge is slipped between the crosshead and the rocker lever socket.

Measure lash by inserting a feeler gauge between the crosshead and the rocker lever socket. If the lash measurement is out of specification, loosen the locknut, and adjust the lash to nominal specifications.



Lash Specifications		
	mm	**in**
Intake	0.254	0.010
Exhaust	0.508	0.020

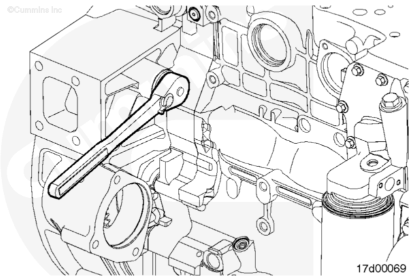
Tighten the locknut and remeasure.

Torque Value: 24 n•m [212 in-lb]

⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure the timing pin is disengaged.

Using barring tool, Part Number 3824591, rotate the crankshaft 360 degrees.



Following the same steps and specifications as previously stated, measure lash for the following rocker levers:

(E = exhaust, I = Intake)

Four-cylinder 2E, 3I, 4E, and 4I:

Six-cylinder 2E, 3I, 4E, 5I, 6I, and 6E.

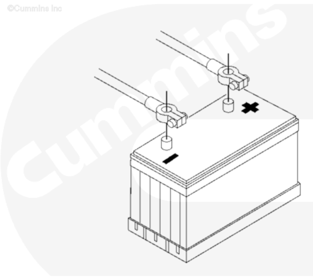
Reset if out of specification.

Finishing Steps

- Install the gasket and rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html).

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



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- Connect the batteries
- Operate the engine and check for leaks.

Last Modified: 24-Apr-2006